

Ryan Carpenter

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SUMMARY

Gameplay Programmer and Technical Artist with experience developing games independently and in collaboration with studios. Strong background in procedurally generated content, realtime rendering pipelines, gameplay systems, multiplayer, VR, and performance optimization. Active in game jams and community driven projects.

EDUCATION

B.S. Computer Science - Arizona State University

May 2025

SKILLS

Languages: C#, C++, Python, Dart

Tools and Pipelines: Unity, VR, NGO, UGS, NUnit, Visual Studio, ASP.NET, Git

Technical Art: Shaders (HLSL, Cg, Shader Graph), Materials, VFX, Blender, Rigging, Adobe Creative Suite

PROJECTS

Labyrinth Algorithm - Individual Project

Nov 2024 - Present

- Interconnected 3D procedural dungeon generator package made in Unity/C#. Applies graph theory, Delaunay triangulation, A*, and greedy algorithms to produce navigable, non-repeating layouts with some linearity.
- Verified algorithm robustness when handling edge cases by implementing unit tests in Unity's Testing Framework and NUnit.
- Reduced generator's CPU overhead by 25% through profiling and optimization.
- Created editor tools and reusable libraries to improve scalability, usability, and setup speed.

Oddinary Farm - Team Project

May 2025 - July 2026

- Engineered client-authoritative multiplayer systems in Unity, including networked combat synchronization, environmental shaders, profiling, and player progression systems.
- Improved gameplay feed and visual feedback by designing and implementing systems integrated with animation and VFX.
- Shipped playtest on Steam with over 400 wishlists and won an award at Kaedim Indie Showcase at GDC for best presentation.

Unstable Star - Team Project

Sept 2023 - May 2024

- Shipped a narrative-driven bullet hell, taking initiative of the project from initial concept through a full public release.
- Led a year-long project as the systems lead of my university's VGDC with 20+ student developers. Oversaw design and systems architecture from concept through release.

EXPERIENCE

Software Project Coordinator, Go Together Inc. - Remote

Aug 2025 - Present

- Led Agile development efforts between stakeholders and a four-person engineering team to successfully deliver the UI/UX (Figma) and feature implementation of a navigation-based mobile application released on App Store and Google Play.
- Led software QA with developers performing unit testing and cognitive walkthroughs, ticketing 400+ bugs.

Studio Associate (Unity XR), ASU Enterprise Technology (Next Lab) - Tempe, AZ

Oct 2023 - June 2025

- Developer of a cross-platform (desktop & VR) career explorer with simulated job-based minigames. Independently built three fully polished minigames (pharmacy, manufacturing, urban planning) within a shared Unity codebase.
- Developer of a VR life-like stroke scale simulator, for clinical education. In charge of building an interactive physics system and voice recognition model.